

Version 7.0 — Data Ingestion, Cleaning, Merging, Validation & Variable Preprocessing

Full engineering specification for the parametric-temporal-windowing pipeline with PopulationSim/IPU synthetic population integration

Project: Evaluating Emerging AV Congestion, Safety, and Modal Shift Dynamics Using SFCTA's SF-CHAMP and Dynamic Network Simulation Framework

Division: Strategy and Technology Policy, San Francisco County Transportation Authority (SFCTA)

Document basis: Reconstructed from `AV_SFCTA_Policy_Project_Spec_v7.txt` (internal version label 6.3-PRODUCTION), cross-referenced against v6 full-detail equations, the Integrated Methodology Primer, and the current `backend/` reference implementation

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Status: Engineering reconstruction — v7's source document carries Sections 4-5 as headers only; this document restores full implementable detail

Purpose of this document. The original v7 specification is a valid architectural document but is not directly buildable: two of its methodological sections are headers with no equations or thresholds, several ETL rules are stated in prose without pseudocode, and no single table consolidates the constants and thresholds an engineer would need to configure the pipeline. This document reconstructs v7 end-to-end — ingestion through variable preprocessing — at implementation grade, names a system of record ("source of truth") for every data element used downstream, consolidates every numeric constant and its justification, and documents the failure modes actually observed when this spec was implemented in the project's reference codebase.

Table of Contents

1. Scope & Document Conventions	3
2. Data Source Inventory & Systems of Record	3
3. Ingestion Layer Specification	4
4. Cleaning & Harmonization (ETL Rules)	6
5. Merging & Entity Resolution	8
6. Validation — QA Gateway Suite (Gates 1-6)	9
7. Variable Preprocessing / Feature Engineering	11
8. Consolidated Constants & Thresholds Reference	13
9. Common Implementation Errors (Observed)	14
10. Methodology Caveats	15
11. Appendix — Phase-to-Section Crosswalk	16

1. Scope & Document Conventions

This specification covers the full v7 pipeline as it must run to satisfy the six-gate QA suite defined in the v7 spec: ingestion of six external data sources, three-plus-one ETL cleaning rules, entity resolution/merging across incident logs and demographic seed data, PopulationSim/IPU synthetic population construction, and preprocessing of every input variable consumed by the nested-logit mode-choice model, the HCM incident-capacity model, and the Bayesian SPRT safety engine.

v7's headline addition over v6 is **PopulationSim (ActivitySim) integration** to operationalize the `Income_Group` categorical variable — this is treated as first-class throughout this document, including the QA Gate 6 threshold it introduces.

Convention	Meaning
Constant/threshold marker HARD	Pipeline halts / rejects the run if violated. No downstream stage may consume the output.
Constant/threshold marker SOFT	Logged as a warning; run proceeds but the artifact must be flagged in the QA report.
[START_DATE, END_DATE]	Inclusive snapshot window. All temporal filters bind to this pair; see §3.0.
Green callout	Source-of-truth declaration — which system is authoritative for a given data element, and (where relevant) which system is explicitly <i>not</i> authoritative.
Red callout	Implementation error observed in the current reference codebase — included so it is not repeated.
Yellow callout	Methodology caveat — a statistical or data-validity limitation that survives correct implementation.

1.1 Global Temporal Parameters

Every ingestion, cleaning, QA, and modeling stage inherits two mandatory parameters, plus one derived value and one derived control window:

```
START_DATE (YYYY-MM-DD) — inclusive lower bound
END_DATE (YYYY-MM-DD) — inclusive upper bound
Delta_Days = END_DATE - START_DATE + 1
CONTROL_START = START_DATE - 21 days    # pre-period baseline for Causal Proxy 2
```

2. Data Source Inventory & Systems of Record

Every downstream field must trace to exactly one system of record. Where two sources overlap (most importantly DMV vs. CPUC for AV VMT), the spec must state which one governs — ambiguity here is the single most common source of silently-wrong denominators in this class of pipeline.

Source	System of record for	Access	Refresh	Explicit exclusions
DataSF SODA API	SF 311 cases, city GIS layers	REST (Socrata), <code>\$where</code> filter	Continuous	—
NHTSA SGO	Federally-reportable ADS/L2 crash incidents (VIN, timestamp, automation category, injury severity)	Scheduled bulk CSV releases (10-day update cycle)	10-day cycle	Not a live queryable REST endpoint for SGO-specific incident records — see §9.3
511.org GTFS-RT	Real-time transit vehicle position & speed (SFMTA)	Protocol Buffers over HTTP, token auth	Sub-minute	—
CPUC Quarterly Filings (via UC Berkeley TIMS)	Commercial passenger-service VMT denominator — the only valid denominator for commercial safety-rate and deadheading calculations	Portal harvest, quarterly	Quarterly	—
DMV Filings (via TIMS)	Testing VMT (with/without safety driver) only	Portal harvest	Quarterly	Explicit Limitation Notice: DMV VMT CANNOT be used as denominator for commercial passenger-service evaluation
OSMnx / OpenStreetMap	Road network topology, edge geometry	<code>osmnx.graph_from_place</code>	Once per snapshot, cached	—
Census ACS PUMS (5-yr)	Household/person microdata <i>seed</i> for PopulationSim (joint distribution of characteristics)	Census API	Annual vintage	Not a source of TAZ-level geography (PUMA only) — see §5.2
ACS Summary File	TAZ-level demographic <i>marginal control</i> totals for IPU expansion	Census API	Annual vintage	—
SFCTA TAZ shapefile	Zone geography for all TAZ-indexed outputs	Static internal GIS asset	Rarely updated	—
SF OpenData — HIN & EPC layers	High-Injury Network flags, Equity Priority Community status per Census Block Group	DataSF portal	5-yr rolling / annual ACS	—

SOURCE OF TRUTH — VMT DENOMINATOR

Commercial AV passenger-service VMT (used in the Bayesian SPRT safety-rate denominator and the Proxy-1 deadheading ratio) is governed exclusively by **CPUC quarterly commercial filings** overlapping `[START_DATE, END_DATE]`, augmented by regional synthetic activity model scaling where filings lag the snapshot window. DMV testing-VMT filings must never be substituted or blended in — they measure a structurally different population of miles (supervised/unsupervised testing, not commercial passenger service).

3. Ingestion Layer Specification

3.1 DataSF SODA API

```
GET https://data.sfgov.org/resource/<dataset-id>.json
$where = requested_datetime >= 'START_DATE T00:00:00' AND requested_datetime <= 'END_DATE T23:59:59'
$limit = 50000    # paginate — see §9.1
$offset = <running offset>
```

```
def fetch_datasf_soda(start_date, end_date, dataset_id, page_size=50000):
    url = f"https://data.sfgov.org/resource/{dataset_id}.json"
    where = (f"requested_datetime >= '{start_date}T00:00:00' "
            f"AND requested_datetime <= '{end_date}T23:59:59'")
    records, offset = [], 0
    while True:
        params = {"$where": where, "$limit": page_size, "$offset": offset}
        resp = _get_with_backoff(url, params)          # retry on HTTP 429/5xx, exp. backoff
        batch = resp.json()
        if not batch:
            break
        records.extend(batch)
        offset += page_size
    return records
```

Parameterization: the `$where` clause must be built with the Socrata client's parameter binding, never raw f-string interpolation into a query string that reaches the network layer unescaped — see error E-2 in §9.

3.2 NHTSA SGO — Incident Telemetry

Ingested as scheduled bulk-file pulls against NHTSA's published Standing General Order crash/incident report releases, filtered post-download to `incident_date BETWEEN START_DATE AND END_DATE`. This is a file-drop cadence (10-day update cycle per the SGO reporting requirement), not a live poll — see error E-3 in §9 for the wrong-endpoint failure mode this replaces.

3.3 GTFS-Realtime / 511.org

```
def poll_gtfs_rt(agency="SF", token=SF511_TOKEN, interval_sec=25):
    """Persistent poller — GTFS-RT overwrites in place; a one-shot
    request captures a single instant, not a time series."""
    while True:
        pb = fetch_protobuf(f"http://api.511.org/transit/VehiclePositions"
                           f"?api_key={token}&agency={agency}")
        write_to_timeseries_store(parse_gtfs_rt(pb), ts=now())
        sleep(interval_sec)
```

SOURCE OF TRUTH — TRANSIT GROUND TRUTH

GTFS-RT breadcrumbs, contrasted against the static GTFS schedule, are the sole source of truth for `Actual_Travel_Time` and the delay vector used in both Causal Proxy 2 and QA Gate 3 (v8) / downstream skim validation (v7). Static GTFS alone is a schedule, not an observation.

3.4 CPUC / DMV via UC Berkeley TIMS

Harvested for overlapping quarterly bounds; both series are ingested (DMV and CPUC) but tagged at ingestion time with a `vmt_class` field (`testing` vs. `commercial`) so the Section 2 exclusion rule is enforceable downstream by a single filter rather than by tribal knowledge.

3.5 OSMnx Road Network Graph

```
def ingest_osmnx_graph(place="San Francisco, California, USA", crs="EPSG:26910"):
    G = ox.graph_from_place(place, network_type="drive")
    G_proj = ox.project_graph(G, to_crs=crs)
    graph_hash = hash_graph(G_proj)
    if graph_hash == _last_cached_hash():
        return _load_cached(G_proj)          # skip re-download if OSM unchanged
    _cache(G_proj, graph_hash)
    return G_proj
```

3.6 ACS PUMS + TAZ Marginal Controls (PopulationSim inputs)

Input	Content	Geography
Seed table	Household/person joint-distribution microdata	PUMA
Marginal controls	Aggregate household-count, income-tier, worker, vehicle-ownership targets	TAZ
Crosswalk	PUMA → TAZ correspondence	Population-weighted areal apportionment via Census block — see §5.2

4. Cleaning & Harmonization (ETL Rules)

4.1 Rule 1 — Deduplication

Cross-references SGO, 311, and SFFD logs to collapse independent reports of the same physical incident into one `Episode_ID` (structurally identical to healthcare claims deduplication, per the Integrated Primer — overlapping reports of one clinical episode collapsed via deterministic keys).

```
def apply_rule_1_deduplication(sgo_gdf, time_tol_min=5, spatial_tol_m=50):
    sgo_gdf = sgo_gdf.sort_values("timestamp")
    sgo_gdf["geometry_buf"] = sgo_gdf.geometry.buffer(spatial_tol_m)
    candidates = gpd.sjoin(sgo_gdf.set_geometry("geometry_buf"), sgo_gdf,
                           predicate="intersects", lsuffix="a", rsuffix="b")
    same_vin = candidates["vin_a"] == candidates["vin_b"]
    near_time = (candidates["timestamp_a"] - candidates["timestamp_b"]).abs() \
        <= pd.Timedelta(minutes=time_tol_min)
    dupes = candidates[same_vin & near_time]
    episode_map = _union_find_episodes(dupes) # connected components -> Episode_ID
    sgo_gdf["episode_id"] = sgo_gdf.index.map(episode_map)
    return sgo_gdf.sort_values("timestamp").drop_duplicates("episode_id", keep="first")
```

COMPOSITE KEY

VIN (exact) AND timestamp (± 5 min) AND spatial radius (50 m) — all three must match for two records to be collapsed. Matching on VIN+timestamp alone (see error E-1, §9) will fail to catch same-incident reports filed by a technology developer and a ride-hail operator under the SGO 30-second rule, since their independently-logged timestamps commonly differ by more than a naive exact-match tolerance.

4.2 Rule 2 — Spatial & Temporal Boundary Enforcement

```
KEEP row WHERE START_DATE <= timestamp <= END_DATE (inclusive)
AND ST_Within(geometry, SF_COUNTY_BOUNDARY)
```

```
def apply_rule_2_boundary(df, ts_col, geom_col, start_date, end_date, county_boundary):
    df[ts_col] = pd.to_datetime(df[ts_col])
    start_dt = pd.to_datetime(start_date)
    end_dt = pd.to_datetime(end_date) + pd.Timedelta(hours=23, minutes=59, seconds=59)
    in_time = df[ts_col].between(start_dt, end_dt)
    in_space = df[geom_col].within(county_boundary)
    return df[in_time & in_space]
```

4.3 Rule 3 — GTFS Anomaly Cleaning

```
DROP row WHERE speed_mph > 65.0 OR distance_jump_m_per_sec > 500.0
```

Applied prior to the "early skip" trajectory-halting step described in the Primer's telemetry QA section — anomalies must be dropped before trajectory imputation, not after, or a single GPS jump will corrupt the interpolated path on both sides of it.

4.4 Rule 4 — Spatial/CRS Harmonization (named in the Primer, absent from v7's ETL rule list)

All ingested API data arrives in mixed or absent coordinate reference systems. This rule is implied by the Primer's QA discussion but not enumerated in v7's own Rule 1-3 list — it is added here as Rule 4 because every spatial join in Rules 1 and downstream in §5 silently depends on it.

```
def apply_rule_4_crs_harmonization(gdf, target_crs="EPSG:26910"):
    if gdf.crs is None:
        gdf = gdf.set_crs("EPSG:4326") # assume WGS84 for unlabeled API output
    return gdf.to_crs(target_crs) # NAD83 UTM Zone 10N — metric distances required
                                   # for the 50m/500m thresholds used elsewhere
```

WHY THIS IS RULE 4 AND NOT OPTIONAL

The 50 m deduplication buffer (Rule 1), the 500 m/sec GTFS jump threshold (Rule 3), and the 50 m Causal-Proxy-2 buffer (§7.3) are all metric distances. If a join executes against unprojected WGS84 degrees, or against inconsistently-projected inputs, these thresholds silently evaluate against the wrong units with no error thrown — one of the most common and hardest-to-detect failure classes in spatial ETL. Rule 4 must run before Rules 1 and 3, not after.

5. Merging & Entity Resolution

5.1 Incident-Log Merge (SGO × 311 × SFFD)

Executed as the union-find pass in §4.1; the merged `Episode_ID` becomes the primary key for QA Gate 1 (telemetry cross-reconciliation).

5.2 PUMA → TAZ Crosswalk (PopulationSim geography)

ACS PUMS seed data is published at PUMA geography; PopulationSim requires TAZ-level marginal controls and produces a TAZ-indexed synthetic population. PUMAs are larger than, and not aligned to, TAZ boundaries — a naive centroid-in-polygon join misallocates population near PUMA/TAZ edges.

```
def build_puma_taz_crosswalk(puma_shp, taz_shp, block_pop_table):
    # 1. Intersect PUMA and TAZ polygons at the Census block level
    blocks = gpd.overlay(block_pop_table, puma_shp, how="intersection")
    blocks = gpd.overlay(blocks, taz_shp, how="intersection")
    # 2. Apportion each block's population by area share across split TAZ fragments
    blocks["frac"] = blocks.geometry.area / blocks.groupby("block_id").geometry.area.transform("sum")
    blocks["apportioned_pop"] = blocks["population"] * blocks["frac"]
    # 3. Aggregate to PUMA x TAZ population-weighted crosswalk
    xwalk = blocks.groupby(["puma_id", "taz_id"])["apportioned_pop"].sum().reset_index()
    xwalk["weight"] = xwalk["apportioned_pop"] / xwalk.groupby("puma_id")["apportioned_pop"].transform("sum")
    return xwalk    # weight = share of PUMA's population that falls in this TAZ
```

FAILURE MODE THIS AVOIDS

A centroid join assigns 100% of a PUMA's seed households to whichever TAZ contains the PUMA's geometric centroid — for irregularly-shaped PUMAs (common along SF's waterfront and southeast industrial areas) this can misassign entire income strata to the wrong TAZ. Population-weighted block-level apportionment is the standard fix and is required before IPU expansion.

5.3 Household/Person Merge Keys for PopulationSim

Table	Primary key	Foreign key
Seed households (PUMS)	hh_id	puma_id
Seed persons (PUMS)	person_id	hh_id
TAZ marginal controls (ACS SF)	taz_id	joined via crosswalk in §5.2
Synthetic output	synth_hh_id	taz_id, hh_id (expansion source)

IPU Convergence Criterion

```
STOP WHEN max(|weight_iter_n - weight_iter_(n-1)|) / weight_iter_(n-1) < 0.0001
OR iteration_count >= 100    # hard cap — treat as non-convergence, not success
```

6. Validation — QA Gateway Suite (Gates 1-6)

Every gate below must return a boolean pass/fail *and* the underlying statistic — a bare `True` with no supporting value (as currently returned by the reference `qa_gates.py` stubs) is not an acceptable implementation of any gate in this suite.

Gate	Name	Statistical logic	Threshold	On failure
1	Telemetry Cross-Reconciliation	VIN-level set difference: NHTSA SGO VINs not present in CPUC log for the window	0 missing events HARD	Halt — surface the missing VIN list, do not proceed to modeling
2	SF-CHAMP Skim Convergence	RMSE between DTA output skims and SF-CHAMP input skims across iterations; utility-bound check on <code>mu_FH</code>	$RMSE < 0.01$ AND $0 < \mu_{FH} \leq 1.0$ HARD	Continue DTA↔SF-CHAMP loop up to iteration cap; halt if cap reached unconverged
3	Proxy 1 Ground-Truth Calibration	MAPE of vision-based deadheading estimate vs. CPUC audit baseline, matching snapshot months	$MAPE \leq 12\%$ HARD	Reject Proxy 1 output for the window
4	Proxy 2 Placebo Validation	Synthetic-control placebo test on non-incident corridors during the window	placebo $p > 0.10$ HARD	Reject Causal Proxy 2 delay estimate
5	SPRT Stress Test	10,000 Monte Carlo low-VMT (<100k) draws; measure false-demotion rate against 1.5× human rate threshold	false demotion rate $< 1.0\%$ HARD	Do not activate Bayesian reversion engine for this snapshot config
6 (v7 new)	Demographic Marginal Validation	MAPE of PopulationSim synthetic counts vs. ACS TAZ marginals, by income tier	$MAPE \leq 1.0\%$ HARD (see caveat)	Reject synthetic population; do not feed <code>Income_Group</code> into nested logit

6.1 Reference Implementations

```
def qa_gate_1_telemetry_reconciliation(nhtsa_df, cpuc_df):
    missing = set(nhtsa_df["vin"]) - set(cpuc_df["vin"])
    return len(missing) == 0, missing

def qa_gate_3_proxy1_calibration(vision_estimate, cpuc_baseline):
    mape = (abs(vision_estimate - cpuc_baseline) / cpuc_baseline).mean()
    return mape <= 0.12, mape

def qa_gate_4_placebo(placebo_p_values):
    worst_p = min(placebo_p_values) # most damaging placebo result governs
    return worst_p > 0.10, worst_p

def qa_gate_6_demographic_marginal(synth_by_taz_income, acs_marginal_by_taz_income, min_cell_n=30):
    eligible = acs_marginal_by_taz_income[acs_marginal_by_taz_income >= min_cell_n].index
    ape = (synth_by_taz_income[eligible] - acs_marginal_by_taz_income[eligible]).abs() \
          / acs_marginal_by_taz_income[eligible]
    return ape.mean() <= 0.01, ape.mean()
```

GATE 6 CELL-SIZE FLOOR

1.0% MAPE is a tight bar for small TAZ × income-tier cells — relative error is mechanically inflated at low counts. The `min_cell_n` floor above (only score cells with an ACS control of ≥30 households) prevents the gate from failing on statistical noise in sparsely-populated TAZ/income combinations rather than genuine calibration error. This floor is an addition to v7's stated gate logic, not present in the original spec text.

7. Variable Preprocessing / Feature Engineering

7.1 Nested Logit — For-Hire Nest Inputs

Variable	Derivation	Source
Fare_AV , Fare_H	Expected trip fare (\$) — pulled from CPUC commercial filings where reported per-trip, else imputed from posted per-mile/per-minute operator rates × trip distance/duration from OSMnx skims	CPUC filings + OSMnx skims
IVTT , IVTT_Transit	In-vehicle time (min) from DTA skims (AV/TNC) or GTFS-RT actual travel time (transit)	DTA output / GTFS-RT
OVTT , Wait_Time	Out-of-vehicle wait/pickup time (min) — GTFS-RT headway-derived for transit; operator-reported pickup ETA distributions for AV/TNC	GTFS-RT / CPUC
Comfort_Discount	Binary/scaled index — not directly observed in any ingested source ; requires an explicit assumption or survey-transfer value (flag as unestimated parameter, \$10)	Assumed — no source of truth
Income_Group	Categorical tier assigned per synthetic household from PopulationSim output (\$5-6)	PopulationSim synthetic population

7.2 Income-Segmented VOT

```
VOT_k = b_ivtt_k / b_fare_k    # $/hour, computed per income segment k
```

Requires segment-specific `b_ivtt_k` and `b_fare_k` — in v7 these remain assumed constants (not empirically estimated; that shift is v8's contribution). Preprocessing step: bin PopulationSim's continuous household income into the same k tiers used by the EPC definition so `VOT_k` aligns to Equity Priority Community reporting, not an arbitrary quantile split.

7.3 Causal Proxy Pipeline Inputs

Proxy 1 — Deadheading VMT strata weights

```
R_deadhead_z = (VMT_Total_Observed_z - VMT_Passenger_Serviced_z) / VMT_Total_Observed_z
R_deadhead_Snapshot = Σ w_z · R_deadhead_z , z ∈ {CBD Corridors, Residential Arterials, Staging Hubs}
```

Preprocessing requirement not specified in the source spec: `w_z` (spatial weighting factor) must be computed as each stratum's share of total observed AV VMT in the window — `w_z = VMT_Total_Observed_z / Σ VMT_Total_Observed` — and recomputed per snapshot, not hardcoded, since fleet deployment shifts stratum share over time.

Proxy 2 — Spatio-temporal buffer

```
Delay_AV_Net = t_actual(t_incident) - t_baseline(t_incident - 21d to t_incident - 1d)
Match window: transit corridor within 50 m of incident, ±15 min of t_incident
```

7.4 HCM Non-Linear Capacity Drop Inputs

```
C_effective = C_base · [1 - (1.1 · (N_blocked/N_total)^0.8 + F_rubberneck)]
```

Input	Preprocessing rule
N_blocked , N_total	Derived from OSMnx edge lane-count attribute joined to incident location; default to conservative 2-lane assumption with a logged warning if lane count is missing from OSM tags
F_rubberneck	Clipped to <code>[0.05, 0.25]</code> — any computed/assumed value outside this range must be clamped, not passed through

7.5 Bayesian SPRT Prior Specification

```
theta ~ Gamma(alpha_0, beta_0)
theta | Snapshot ~ Gamma(alpha_0 + K_snapshot, beta_0 + VMT_snapshot)
```

`alpha_0` , `beta_0` must be set from the ODD-matched historical human-driver baseline for the matching Road Class × Speed Limit × Time-of-Day stratum — not a flat uninformative prior — or the posterior will be dominated by whatever prior an engineer happened to hardcode (see error E-6, §9).

8. Consolidated Constants & Thresholds Reference

Single source of truth for every numeric constant used anywhere in the v7 pipeline. If a value in code does not match this table, the code is wrong, not the table.

Constant	Value	Applies to	Basis / justification
Dedup timestamp tolerance	±5 min	Rule 1	SGO 30-second filing rule + operator reporting lag margin
Dedup spatial tolerance	50 m	Rule 1, Proxy 2	Primer / v6 spec — corridor-matching radius
GTFS speed anomaly ceiling	65 mph	Rule 3	Physically implausible for SF surface transit
GTFS distance-jump ceiling	500 m/sec	Rule 3	GPS/telemetry glitch signature
Target CRS	EPSG:26910 (NAD83 UTM 10N)	Rule 4, all spatial joins	Metric projection required for meter-based thresholds
Control window offset	21 days pre-incident	Proxy 2 baseline	3-week pre-incident baseline, v6/v7 spec
Proxy 2 temporal match window	±15 min	Proxy 2	v6/v7 spec §4.2
Gate 1 tolerance	0 missing events	QA Gate 1	v6/v7 spec §6.3 — zero-tolerance by design
Gate 2 skim RMSE ceiling	< 0.01	QA Gate 2 / DTA↔SF-CHAMP loop	v6/v7 spec §3.3
μ_{FH} bounds	$0 < \mu_{FH} \leq 1.0$	QA Gate 2 / nested logit validity	Discrete-choice theory — nest scale-parameter constraint
Gate 3 MAPE ceiling	12%	QA Gate 3 / Proxy 1	v6/v7 spec §6.3
Gate 4 placebo p-floor	> 0.10	QA Gate 4 / Proxy 2	v6/v7 spec §6.3
Gate 5 Monte Carlo draws	10,000	QA Gate 5 / SPRT stress test	v6/v7 spec §6.3
Gate 5 low-VMT band	< 100,000 VMT	QA Gate 5	v6/v7 spec §6.3
Gate 5 false-demotion ceiling	< 1.0%	QA Gate 5	v6/v7 spec §6.3
Gate 6 MAPE ceiling	1.0%	QA Gate 6 / PopulationSim	v7 spec §6.3 — see cell-size floor caveat, §6.1
Gate 6 min cell size	$n \geq 30$ households	QA Gate 6	Added in this reconstruction — not in original spec
SPRT demotion trigger	$P(\theta_{AV} > 1.5 \cdot \theta_{Human}) > 0.95$	Bayesian reversion engine	v6/v7 spec §5.2
SPRT advancement trigger	$P(\theta_{AV} < \theta_{Human}) > 0.90$ AND $VMT \geq 1,000,000$	Bayesian reversion engine	v6/v7 spec §5.2
Hysteresis window	Rolling 30 days	Bayesian reversion engine	v6/v7 spec §5.3
Post-demotion compliance period	60 consecutive days	Bayesian reversion engine	v6/v7 spec §5.3
HCM rubberneck factor range	[0.05, 0.25]	Incident capacity drop	v6/v7 spec §3.3
IPU convergence tolerance	< 0.0001 relative weight change	PopulationSim	Added in this reconstruction — standard IPU practice
IPU iteration cap	100	PopulationSim	Added in this reconstruction
SODA pagination page size	50,000	Ingestion §3.1	Fixes 1,000-row truncation bug — see E-1

9. Common Implementation Errors (Observed)

Each of the following was found in the project's reference implementation and is documented here so it is not silently reintroduced when v7 is rebuilt to this spec.

E-1 — SILENT PAGINATION TRUNCATION (DATASF SODA)

The reference ingestion script calls the SODA endpoint with `$limit=1000` and no offset loop. Socrata's default/observed page cap silently returns only the first 1,000 records for any window with more activity than that — no error is thrown, so a snapshot can under-count 311/incident volume with no visible failure signature. Fix: paginate as shown in §3.1.

E-2 — UNPARAMETERIZED QUERY CONSTRUCTION

Building the SODA `$where` clause via f-string interpolation of `start_date / end_date` directly into the request is an injection surface if those values are ever sourced from anything less trusted than a CLI flag (e.g., a future web form). Use the client library's parameter binding.

E-3 — WRONG NHTSA ENDPOINT

The reference code calls NHTSA's general `CrashAPI/GetCaseList` endpoint, which serves general crash statistics, not ADS/SGO-specific incident reports. The correct source of record is NHTSA's published Standing General Order report releases (bulk files), per §3.2 — the general Crash API will not contain the automation-category and VIN fields Rule 1 dedup depends on.

E-4 — ONE-SHOT GTFS-RT FETCH TREATED AS A TIME SERIES

GTFS-RT is a transient, overwrite-in-place feed. A single request captures one instant. Reference code that calls the endpoint once per script invocation cannot reconstruct actual travel time or delay vectors — a persistent poller (§3.3) writing to a time-series store is mandatory, not an optimization.

E-5 — NAIVE PUMA→TAZ CENTROID JOIN

Assigning an entire PUMA's synthetic households to the TAZ containing the PUMA's centroid misallocates population for irregularly-shaped PUMAs. Use the population-weighted block-level apportionment in §5.2.

E-6 — FLAT/UNINFORMATIVE BAYESIAN PRIOR HARDCODED WITHOUT ODD STRATIFICATION

Setting `alpha_0 / beta_0` to an arbitrary flat prior rather than the ODD-matched (Road Class × Speed Limit × Time-of-Day) human baseline understates the informativeness the spec's own decision-trigger thresholds assume — the 0.95/0.90 posterior-probability triggers in §7.5 were calibrated against an informative prior, not a flat one.

E-7 — EXACT-MATCH DEDUP INSTEAD OF COMPOSITE-KEY DEDUP

Deduplicating on exact `(vin, timestamp)` equality (rather than the ± 5 min / 50 m composite key in §4.1) will fail to catch the primary target case the rule exists for: two operators independently filing reports of the same incident with different timestamps.

E-8 — CRS ASSUMED RATHER THAN VERIFIED

Spatial joins executed without first confirming/setting a CRS on incoming API data (which frequently arrives with no CRS metadata at all) will run against whatever CRS pandas/geopandas defaults to, silently producing wrong distances for every meter-based threshold in this spec. Apply Rule 4 (§4.4) before any spatial join, not after.

10. Methodology Caveats

These survive correct implementation — they are limits of the design, not bugs to fix.

C-1 — PLACEBO TEST INDEPENDENCE ASSUMPTION (GATE 4)

The synthetic-control placebo test assumes independence across non-incident corridors. Transportation network data is spatially and temporally autocorrelated (adjacent corridors share congestion shocks); without block-bootstrapping or spatial-clustering correction, the placebo p-value threshold (> 0.10) will be systematically too easy to clear, understating false-positive risk in Proxy 2.

C-2 — COMFORT_DISCOUNT AND SEVERAL NESTED-LOGIT COEFFICIENTS HAVE NO SOURCE OF TRUTH

Per §7.1, `Comfort_Discount` and the ASCs (`b_0_AV` , `b_0_H`) are assumed constants in v7, not estimated from any ingested dataset. Every downstream logsum, modal-shift estimate, and consumer-surplus calculation inherits this assumption uncritically. This is the specific gap v8 attempts to close via MLE — it is unresolved in v7 as specified.

C-3 — IPU CELL SPARSITY AT GATE 6

Even with the min-cell-size floor added in §6.1, TAZ $\times$ income-tier cells near San Francisco's smaller or less-populated TAZs will have inherently noisy MAPE. A pass at Gate 6 should be read as "no evidence of systematic miscalibration in well-populated cells," not as "the synthetic population is accurate everywhere."

C-4 — VERSION/DOCUMENT TRACEABILITY

This document reconstructs the file internally labeled `Version: 6.3-PRODUCTION` (filename `..._Spec_v7.txt`). The mismatch between filename-based versioning and internal document versioning should be resolved before this spec is cited in any audit trail — "v7" and "6.3" must not be used interchangeably without a mapping note.

C-5 — 21-DAY CONTROL WINDOW VS. DAY-OF-WEEK EFFECTS

The Proxy 2 baseline (`t_baseline` over the 21 days preceding an incident) is not explicitly stratified by day-of-week or special-event calendar. Transit travel time has strong weekday/weekend and event-driven variance; a straight 21-day median can be biased if the window happens to span an atypical period (e.g., a holiday week). Recommend stratifying the baseline by matching day-of-week at minimum.

11. Appendix — Phase-to-Section Crosswalk

Phase	v6/v7 Workflow Description	Sections in this document
1 (Wk 1–2)	Ingestion Pipeline, PostGIS Architecture & Demographics; PopulationSim IPU run; ETL Rules 1–3; QA Gates 1 & 6	\$3 (Ingestion), \$4 (Cleaning), \$5.2–5.3 (PopulationSim merge), \$6 (Gates 1, 6)
2 (Wk 3–5)	Snapshot Proxy Construction & Calibration	\$7.3 (Proxy inputs), \$6 (Gates 3, 4)
3 (Wk 6–8)	SF-CHAMP & DTA Model Snapshot Simulation	\$7.1–7.2 (nested logit inputs), \$7.4 (HCM capacity), \$6 (Gate 2)
4 (Wk 9–10)	Snapshot Safety Evaluation & Policy Sensitivity Testing	\$7.5 (SPRT priors), \$6 (Gate 5)
5 (Wk 11–12)	Visual Dashboards, Deliverables & Code Handoff	Out of scope for this document — see project README, <code>frontend-dashboard/</code> , <code>executive-engine/</code>

Reconstructed implementation specification — AV_SFCTA_Policy_Project, v7. Compiled from `AV_SFCTA_Policy_Project_Spec_v6.txt`, `_v7.txt`, the Integrated Methodology Primer, and reference-implementation review of `backend/{ingestion,cleaning,proxy_pipeline,qa_gates,analytics_engine}.py`. Constants and thresholds are sourced to the original spec text where cited; items marked "added in this reconstruction" are engineering additions not present in v7's source document and should be reviewed before adoption as binding requirements.